

Three LLMs on Four Blackhole Chips

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Project

End-to-end bringup of three modern LLM families on a single 1×4 **Blackhole P150** mesh: dense GQA, MoE with Gated DeltaNet recurrence, and a hybrid sliding/global-attention model. Each model walks from HF reference to a continuous-batching HTTP chat server, gated against a CPU fp32 oracle.

Hardware

Mesh	1 × 4 Blackhole P150 (4 chips)
DRAM	32 GB GDDR6/chip, 404 GB/s stream
On-chip	110 Tensix cores, 1.39 MiB L1/core
Host link	1.4 GB/s tilize-bound

Models Brought Up

Model	Params	Architecture	Status
Qwen3.6-27B	27B	Dense GQA, TP4	shipped
Qwen3.6-35B-A3B	35B	MoE + Gated DeltaNet	$B=1$
Gemma 4 12B IT	12B	Hybrid sliding/global	shipped

Methodology

A staged ladder, each rung gated on cosine vs. a CPU fp32 oracle: single-layer $\rightarrow$ full forward $\rightarrow$ multi-step decode $\rightarrow$ long context $\rightarrow$ trace $\rightarrow$ continuous batching $\rightarrow$ HTTP. Each new model reused $\sim 80\text{--}90\%$ of the prior bringup. Validation is teacher-forced cosine *and* MAD per layer; six of seven silent-failure classes are invisible at decode pos 0 alone.

What Mattered

Three changes account for most of the throughput delta over a naive port.

- **Paged SDPA, HiFi2 fp16 accumulation** — kernel-time win; ⬆️+62% on 27B TP single-stream.
- **Vocab-sharded lm_head, on-device argmax** — a 600 KB host readback collapses to 8 bytes; +8% Gemma 4, +5% 27B.
- **Two-phase trace warmup** — compile every path first, capture all traces back-to-back; otherwise decode JIT corrupts prefill trace memory.

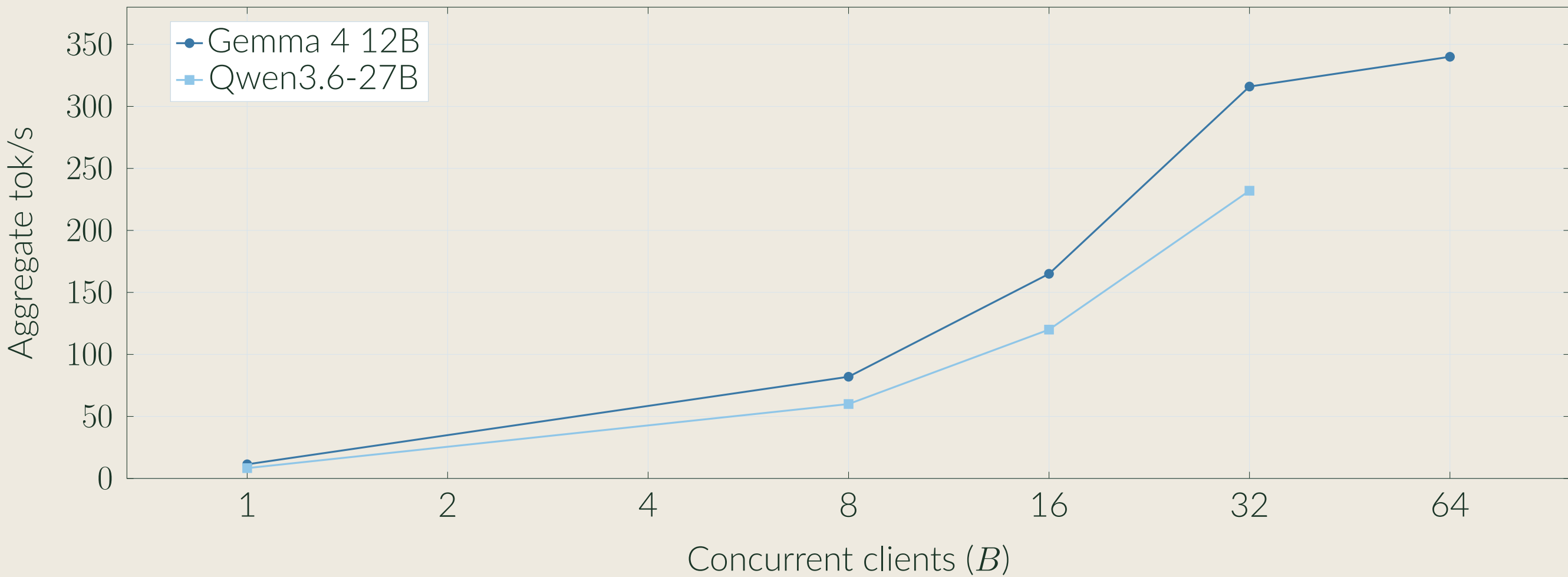
Headline

Gemma 4 12B: 316 tok/s aggregate at 32 clients.

Qwen3.6-27B: 232 tok/s aggregate at 32 clients.

28–29× near-linear scaling from $B=1$ via continuous batching.

Throughput vs. Concurrent Clients



Diminishing returns past $B=32$ (316 $\rightarrow$ 340 at $B=64$).

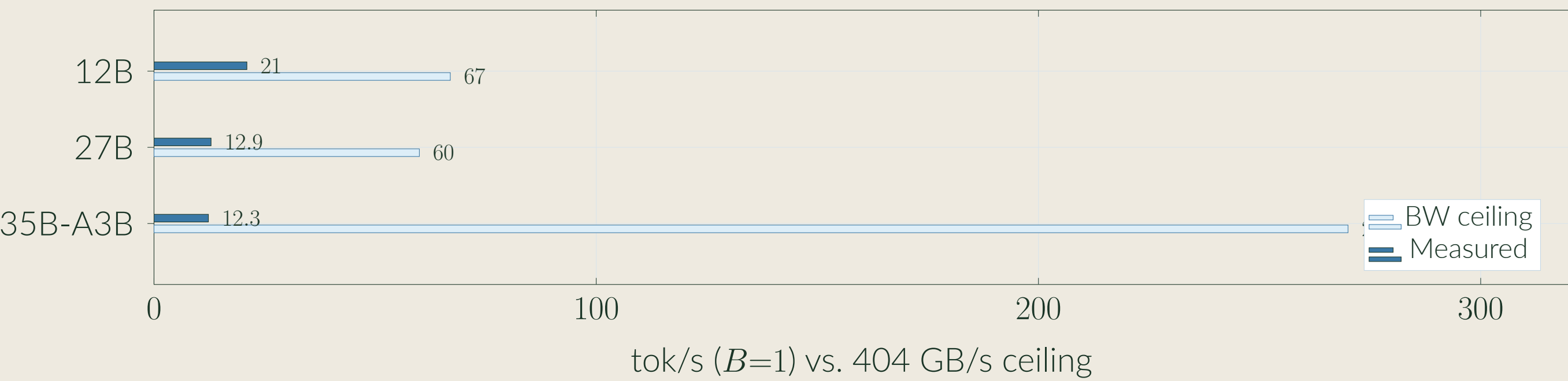
Single-Client Streaming (HTTP, $B=32$ traced)

Model	Prefill	Decode
Gemma 4 12B	78 ms/tok (TTFT 1.40 s)	17.4 tok/s
Qwen3.6-27B	94 ms/tok (TTFT 1.70 s)	11.5 tok/s

Decode is 83% of the $B=1$ dev-harness ceiling.

Roofline: Measured vs. BW Ceiling ($B=1$)

Model	ms/tok	tok/s	% ceiling
Gemma 4 12B	47.5	21.0	⬆️31%
Qwen3.6-27B (TP4)	77.0	12.9	22%
Qwen3.6-35B-A3B	81.2	12.3	4.6%



Correctness

Three orthogonal gates, run per model on every change.

Gate	27B	35B-A3B	Gemma 4
HF cosine ladder	✔️PASS	$\cos_{L_{32}} \geq 0.987$	✔️PASS
Multi-turn KV retention	✔️PASS	3/3 ✔️PASS	✔️PASS
Traced = Eager	✔️PASS	✔️PASS	100/100
Needle haystack	$L=500$	coherent	3/3 ($L \leq 512$)

Validated against a CPU fp32 oracle with per-sub-op hooks. Cosine alone is insufficient ($\cos(a, ka)=1$); MAD is reported at every probed position.

Continuous Batching

Iteration-level Orca-style scheduler: per-step admission/eviction, per-slot paged SDPA over a shared KV pool, slot-level content-keyed prefix cache. Each slot advances at its own position in one batched forward.

Prefix Cache in Practice

Three-turn 27B chat over HTTP (TT_CB_PREFIX_CACHE=1):

Turn	Prompt	Wall	s/tok	Speedup
0 (cold)	36 tok	8.69 s	0.241	—
1 (PC hit)	125 tok	5.84 s	0.047	⬆️5.1×
2 (PC hit)	201 tok	5.92 s	0.030	⬆️8.0×

Per-token cost *drops* as the prompt grows. Metric delta: `pc_hits` +2, `pc_misses` +1.

Conclusion

A single 1×4 Blackhole P150 mesh, with a disciplined bringup recipe, takes a modern LLM from spec to byte-stable HTTP chat at **316 tok/s** on 12B and **232 tok/s** on 27B — letting a small team bring up MoE and hybrid attention without a vendor roadmap.